Final308LoganMoy

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Part 1

#Reading in data and making an altered tibble

```
housedata <-read_csv("https://www4.stat.ncsu.edu/~online/ST308/Data/lbmoy2_house.csv")
```

```
## Rows: 1000 Columns: 13
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (6): Neighborhood, Electrical, RoofStyle, BsmtQual, HeatingQC, LotShape
## dbl (7): SalePrice, BsmtFinSF1, GarageCars, YearBuilt, BsmtUnfSF, YearRemodA...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
housedata2 <- housedata %>% filter(LotShape != "IR2or3"&YearRemodAdd > 1963) %>%
            mutate("SaleinHT" = SalePrice/100000) %>%
  select(-c(BsmtFinSF1, Neighborhood))

kable((housedata2)[1:10,1:6], format = "simple")
```

SalePrice	GarageCars	YearBuilt	BsmtUnfSF	YearRemodAdd	GarageArea
223500	2	2001	434	2002	608
250000	3	2000	490	2000	836
143000	2	1993	64	1995	480
307000	2	2004	317	2005	636
200000	2	1973	216	1973	484
345000	3	2005	177	2006	736
279500	3	2006	1494	2007	840
132000	2	1929	832	2001	576
90000	2	1967	0	1967	516
159000	2	2004	468	2004	576

#Summary statistics of housedata

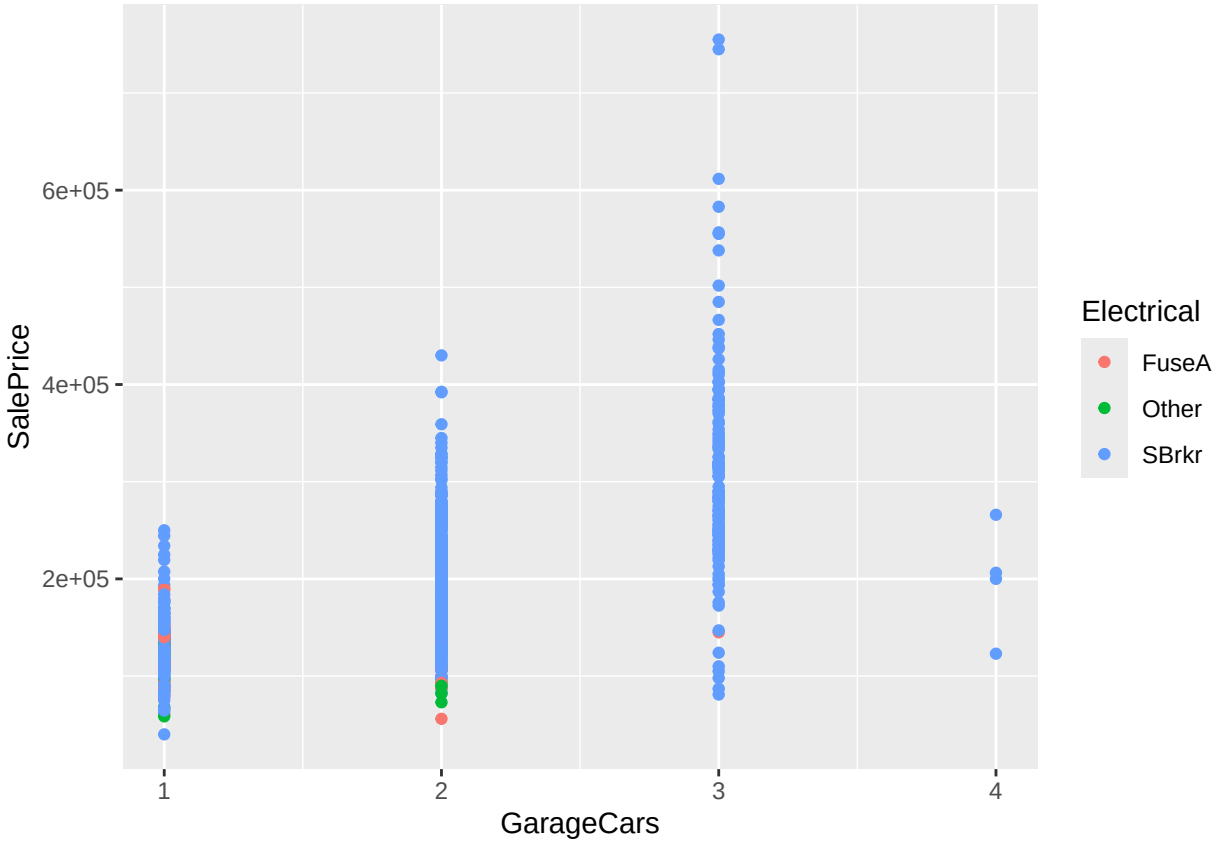
```
sumhd2 <- housedata %>% group_by(Electrical) %>%
          summarize(meanSP = mean(SalePrice), stdSP = sd(SalePrice), quart1SP = quantile(SalePrice,probs
                    meanGC = mean(GarageCars), stdGC = sd(GarageCars), quart1GC = quantile(GarageCars,pr
                    meanYB = mean(YearBuilt), stdYB = sd(YearBuilt), quart1YB = quantile(YearBuilt,probs

kable((sumhd2), format = "simple")
```

Electrical	meanSP	stdSP	quart1SP	quart3SP	meanGC	stdGC	quart1GC	quart3GC	meanYB	st
FuseA	127516.4	30326.68	109900.0	144125	1.303571	0.5016207	1	2	1941.696	18.1
Other	111271.9	30055.47	89250.0	133625	1.400000	0.5026247	1	2	1934.750	17.0
SBrkr	191411.9	81504.71	136903.8	225250	1.923160	0.6153898	2	2	1976.389	28.1

#Creating graphs and models

```
g <-ggplot(housedata, aes(x = GarageCars, y = SalePrice, color = Electrical))
g + geom_point()
```



Most of the Electrical types seem to be SBrkr and fewer are FuseA. Initally there is a mostly positive relationship between Garage car count and sales price, this changes with a count of 4 garage cars (this category has few data points so if there was more data a positive relationship still could be found).

#Creating a MLR model

```
houselm <-lm(SalePrice~ GarageCars + YearBuilt, data = housedata)

summary(houselm)
```

```
##
## Call:
## lm(formula = SalePrice ~ GarageCars + YearBuilt, data = housedata)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -198826  -37765   -8293   25088  482553
##
```

```
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept) -1.252e+06  1.495e+05  -8.372   <2e-16 ***
## GarageCars   6.472e+04  3.607e+03  17.945   <2e-16 ***
## YearBuilt    6.671e+02  7.752e+01   8.605   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 60650 on 997 degrees of freedom
## Multiple R-squared:  0.4383, Adjusted R-squared:  0.4372
## F-statistic:   389 on 2 and 997 DF,  p-value: < 2.2e-16
```

```
predict(houselm, newdata = data.frame(GarageCars = c(2,4), YearBuilt= c(1977, 2000)))
```

```
##        1        2
## 196386.3 341170.8
```

Both of the main effects are significant as both are less than 0.01. There is significant evidence that Garage Cars and Year Built have an effect on the Sale Price.

Part 2 Open Ended

#Old code to read in dataset New data that interested me URL : https://archive.ics.uci.edu/dataset/45/heart+disease Used the processed cleveland data Why it interests me: I just picked this out my old dataset was full of doubles and would not work nicely with the code 1) Which sex (female = 0, male = 1) has higher cholesterol 2) Comparing numeric summaries of resting heart rate (trestbps) vs thalach (maximum hear rate) 3) Number of vessels vs thal

```
heartdata <- read_csv("C:\\Users\\lbmoy2\\Downloads\\heart+disease\\processed.cleveland.data", col_names
  rename("age" = X1,
         "sex" = X2,
         "cp" = X3,
         "trestbps" = X4,
         "chol" = X5,
         "fbs" = X6,
         "restecg" = X7,
         "thalach" = X8,
         "exang" = X9,
         "oldpeak" = X10,
         "slope" = X11,
         "ca" = X12,
         "thal" = X13,
         "num" = X14)
```

Custom Function for data of interest

```
heartfunction = function(data,colname,group = NULL){

  new_data = data[,colname]

  idx = sapply(new_data,is.numeric)
  num_var = new_data[,idx]
  cat_var = new_data[,!idx]

  if (length(num_var) == 0){
```

```
    finallist = c()
  }
  if (length(cat_var) == 0){
    table = "NA"
  }
  if (length(num_var) != 0 ) {
    if (is.null(group )){
      mean <- sapply(num_var, mean)
      var <- sapply(num_var, var)
      quantiles <- sapply(num_var, quantile, probs = c(0.05, 0.95))
      finallist <- list( mean = mean, var = var, quantile = quantiles)
    }


    if (!is.null(group)){ #
      finallist <- data %>% group_by(group) %>% summarize(mean = sapply(num_var, mean), var <- sapply(nu
      }
  }
  if (length(cat_var) != 0 ){
    table <- table(cat_var)
}
  return(list(finallist, table = table))

}
#heartfunction(heartdata, c("chol"), group = heartdata$sex)
heartfunction(heartdata,c("trestbps", "thalach"))
```

```
## [[1]]
## [[1]]$mean
## trestbps  thalach
## 131.6898 149.6073
##
## [[1]]$var
## trestbps  thalach
## 309.7511 523.2658
##
## [[1]]$quantile
##     trestbps thalach
## 5%       108   108.1
## 95%      160   181.9
##
##
## $table
## [1] "NA"
```

```
heartfunction(heartdata,c("ca","thal"))
```

```
## [[1]]
## NULL
##
## $table
##      thal
## ca      ? 3.0 6.0 7.0
##   ?     0   2   0   2
```

```
##   0.0   2 115   8  51
##   1.0   0  29   4  32
##   2.0   0  14   4  20
##   3.0   0   6   2  12
```